

DP Computer Science: A 2.3.1 IP Addressing

1. Core Concepts & Learning Objectives

Learning Intentions

By the end of this lesson, students will be able to:

- Distinguish between **IPv4** and **IPv6** addressing
 - Differentiate between **public** and **private** IP addresses
 - Explain the difference between **static** and **dynamic** IP addresses
 - Describe the role of **NAT** (Network Address Translation) in IP address conservation and network communication
 - Apply understanding of IP addressing to **real-world scenarios**
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Key Vocabulary

Term	Definition
IP Address	A unique identifier for a device on a network
IPv4	The fourth version of IP; uses 32-bit addresses (e.g., 192.168.1.1)
IPv6	The sixth version of IP; uses 128-bit addresses (e.g., 2001:0db8:85a3:0000:0000:8a2e:0370:7334)
Public IP Address	An IP address that is globally unique and accessible over the internet
Private IP Address	An IP address used within a private network (not accessible over the internet)
Static IP Address	An IP address that is manually assigned and does not change
Dynamic IP Address	An IP address that is automatically assigned and may change over time
DHCP	Dynamic Host Configuration Protocol; automatically assigns IP addresses

Term	Definition
NAT	Network Address Translation; translates private IPs to public IPs
Subnet Mask	Identifies which part of an IP address is the network and which part is the host

ATL Skills

Skill	Description
Thinking	Critical thinking, analyzing, evaluating
Communication	Explaining concepts and presenting information
Information Literacy	Researching and gathering information

2. IP Addressing Overview

What is an IP Address?

An IP address is a unique identifier for a device on a network. It allows devices to find and communicate with each other.

Analogy: An IP address is like a **postal address**—it tells the network where to deliver data.

Why Do We Need IP Addresses?

Purpose	Description
Identification	Identifies each device on a network
Location	Indicates where a device is located on the network
Routing	Helps routers forward data to the correct destination
Communication	Enables devices to send and receive data

3. IPv4 vs. IPv6

IPv4

Aspect	Description
Address Size	32 bits
Address Format	Dotted decimal (e.g., 192.168.1.1)
Number of Addresses	Approximately 4.3 billion
Example	192.168.1.1
Status	Depleting (running out of addresses)

IPv4 Structure:

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192 . 168 . 1 . 1

Each octet = 8 bits (0-255)

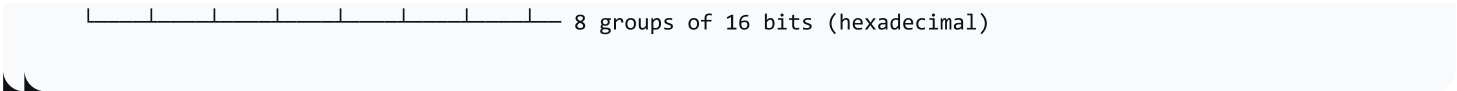
IPv6

Aspect	Description
Address Size	128 bits
Address Format	Hexadecimal with colons (e.g., 2001:0db8:85a3:0000:0000:8a2e:0370:7334)
Number of Addresses	Approximately 340 undecillion (3.4×10^{38})
Example	2001:0db8:85a3:0000:0000:8a2e:0370:7334
Status	Growing (future-proof)

IPv6 Structure:

text

2001:0db8:85a3:0000:0000:8a2e:0370:7334



Comparison Table

Feature	IPv4	IPv6
Address Size	32 bits	128 bits
Number of Addresses	~4.3 billion	~340 undecillion
Format	Dotted decimal	Hexadecimal with colons
Example	192.168.1.1	2001:0db8:85a3::7334
Security	Optional (IPSec)	Built-in (IPSec required)
NAT	Required for address conservation	Not needed
Status	Depleting	Growing

Why Move to IPv6?

Reason	Description
Address Exhaustion	IPv4 addresses are running out
More Addresses	IPv6 provides virtually unlimited addresses
Simplified Header	Faster processing (fewer fields)
Built-in Security	IPSec is mandatory
Better Autoconfiguration	Devices can auto-configure
No NAT	Every device can have a unique public IP

4. Public vs. Private IP Addresses

Public IP Address

Aspect	Description
Definition	Globally unique; accessible over the internet
Assigned By	Internet Service Provider (ISP)
Cost	Usually paid
Example	203.0.113.5
Who Can See It	Anyone on the internet
Use	Web servers, email servers, public-facing devices

Private IP Address

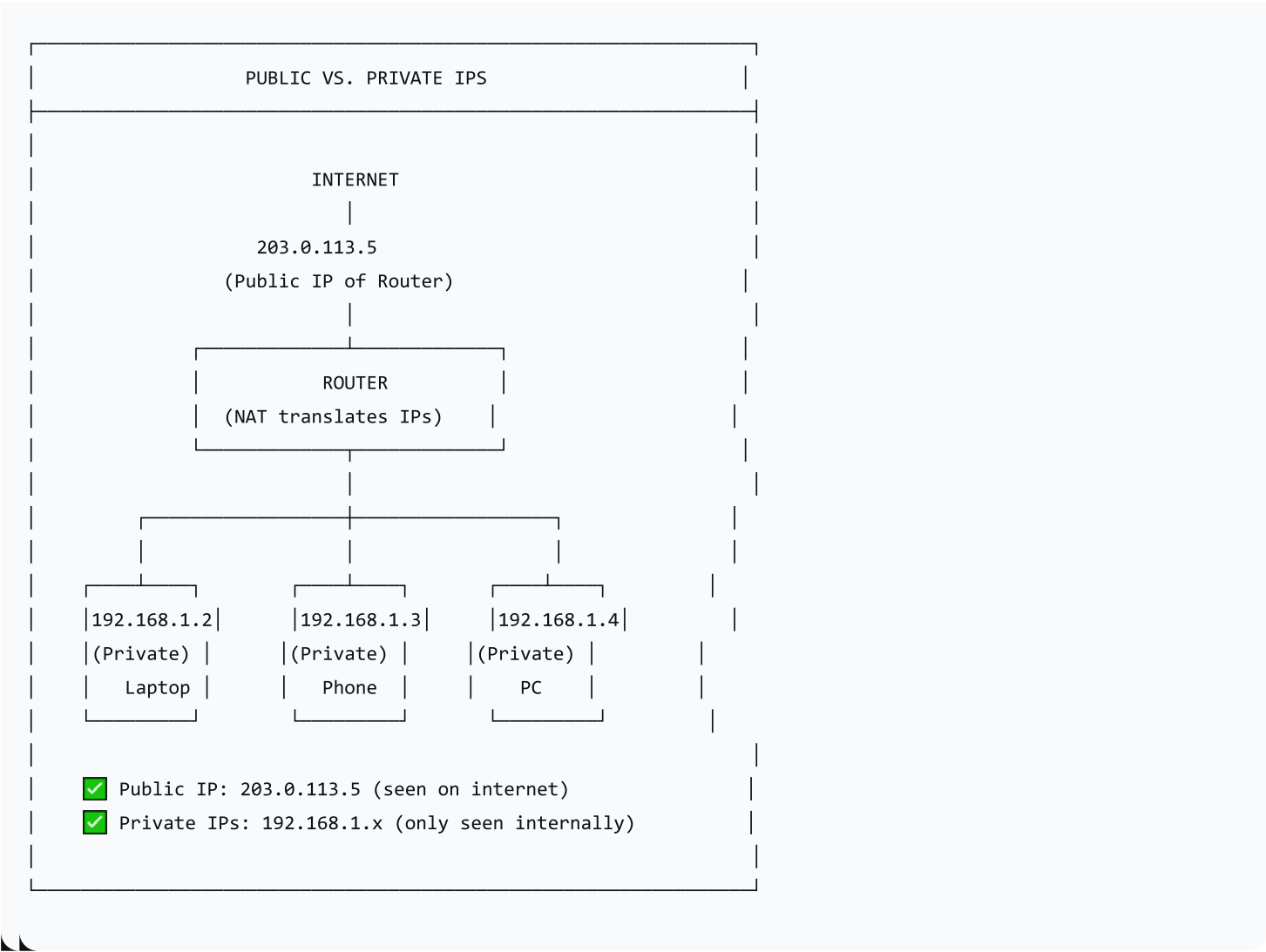
Aspect	Description
Definition	Used within a private network; not accessible over the internet
Assigned By	Router or DHCP server
Cost	Free
Example	192.168.1.1
Who Can See It	Only devices on the same private network
Use	Home networks, office networks, internal devices

Private IP Address Ranges

Range	CIDR	Number of Addresses	Common Use
10.0.0.0 – 10.255.255.255	10.0.0.0/8	16.7 million	Large corporate networks
172.16.0.0 – 172.31.255.255	172.16.0.0/12	1,048,576	Medium-sized networks
192.168.0.0 – 192.168.255.255	192.168.0.0/16	65,536	Home and small office networks

Visual Summary

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5. Static vs. Dynamic IP Addresses

Static IP Address

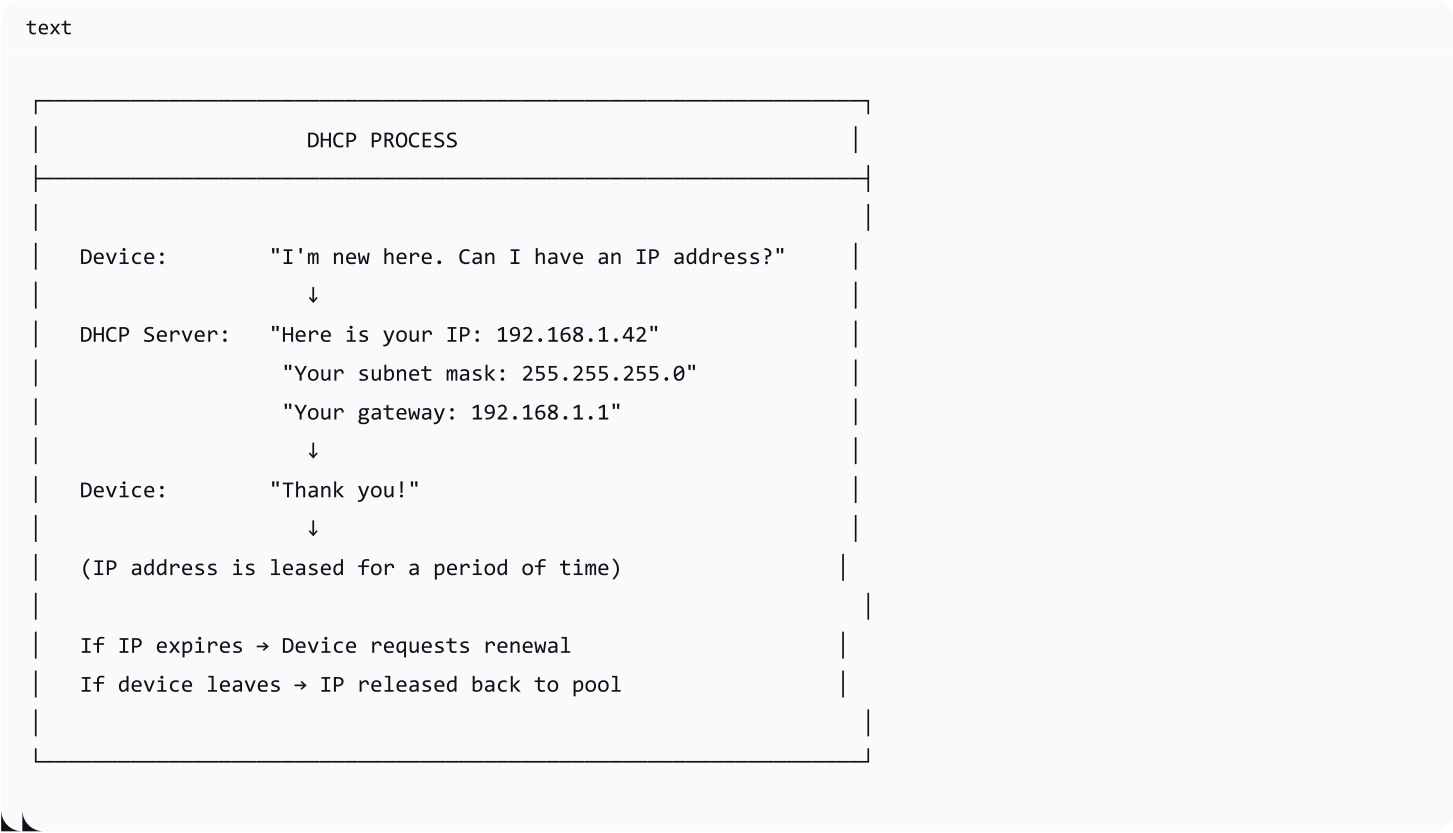
Aspect	Description
Definition	An IP address that is manually assigned and does not change
How It's Assigned	Manually configured on the device
Advantages	Always the same (reliable); easy to find
Disadvantages	Requires manual configuration; less flexible
When to Use	Web servers, email servers, devices that need constant access

Aspect	Description
Example	A company's web server (fixed IP)

Dynamic IP Address

Aspect	Description
Definition	An IP address that is automatically assigned and may change over time
How It's Assigned	DHCP server assigns IP addresses
Advantages	Automatic; flexible; efficient
Disadvantages	IP can change (less reliable for servers)
When to Use	Home computers, smartphones, laptops
Example	Your home Wi-Fi devices

How DHCP Works



Comparison Table

Feature	Static IP	Dynamic IP
Assignment	Manual	Automatic (DHCP)
Changes	Never	May change
Reliability	Very reliable	Depends on lease
Configuration	Requires effort	Effortless
Best For	Servers, printers	Home devices, phones
Cost	Usually higher	Included
Availability	Always available	Available when connected

6. Network Address Translation (NAT)

What is NAT?

NAT (Network Address Translation) translates private IP addresses to public IP addresses, allowing multiple devices to share a single public IP.

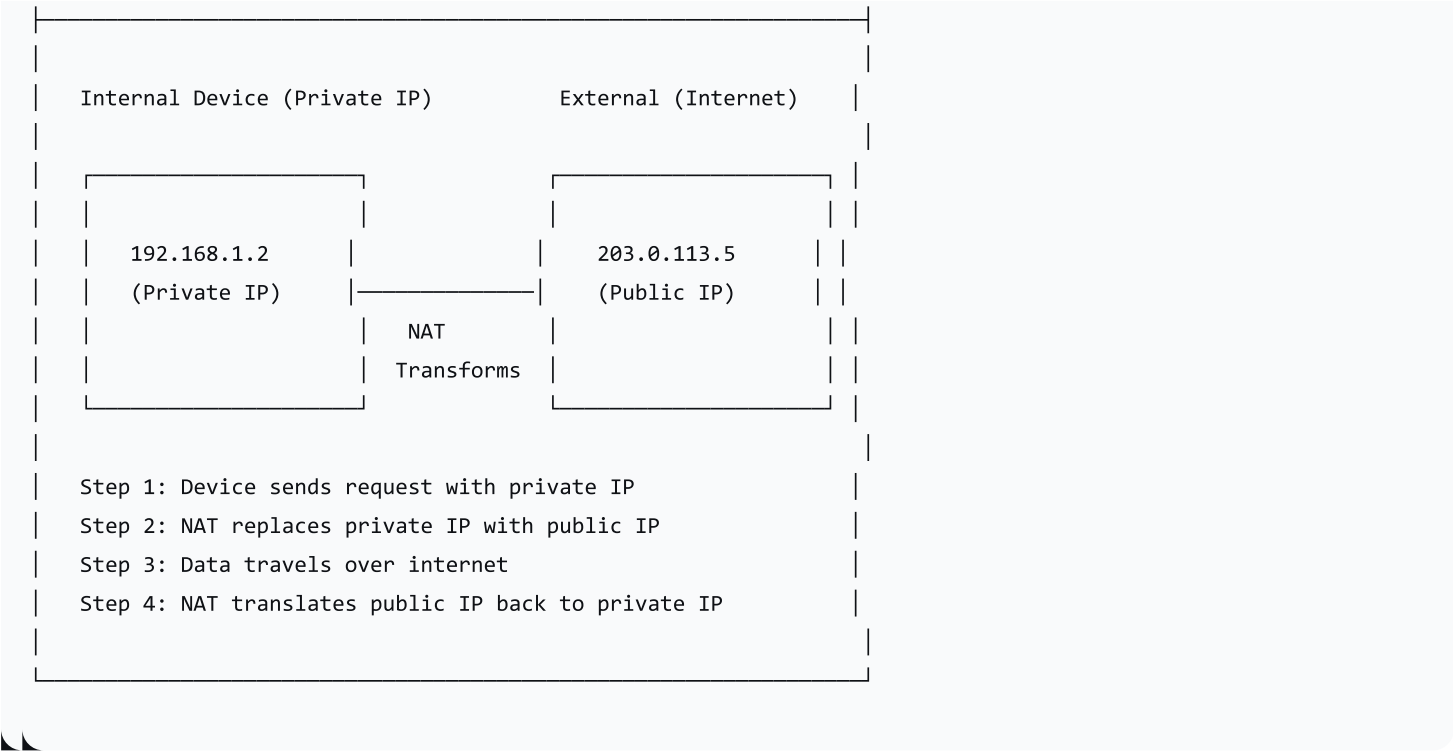
Why is NAT Needed?

Reason	Description
Conserves IPv4 Addresses	Multiple private devices share one public IP
Allows Private Networks	Devices can use private IPs internally
Enhances Security	Hides internal network structure from the internet
Simplifies Configuration	Internal devices don't need public IPs

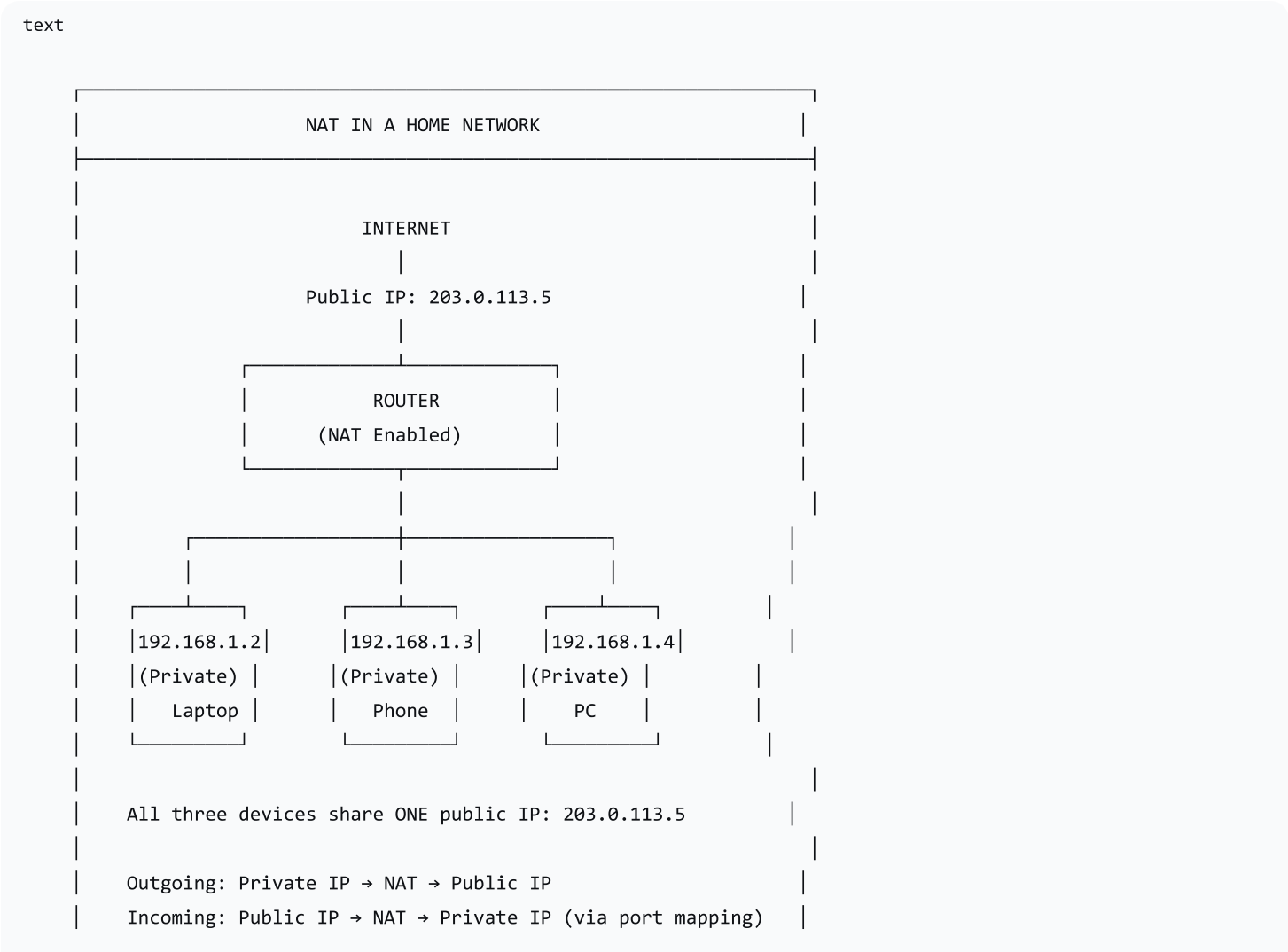
How NAT Works

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HOW NAT WORKS



Visual Example





NAT Benefits

Benefit	Description
Address Conservation	Saves limited IPv4 addresses
Security	Hides internal network structure
Flexibility	Internal IPs can change without affecting external communication
Cost Savings	Only need one public IP for many devices

NAT Limitations

Limitation	Description
No Direct Incoming Connections	Cannot connect directly to internal devices (unless port forwarding is set up)
Complexity	Adds complexity to network design
Performance Overhead	NAT translation adds processing overhead
Not Needed with IPv6	IPv6 eliminates the need for NAT

7. Real-World Scenario: IP Address Types in Action

Scenario: Home Internet

Device	IP Type	How Assigned	Notes
Router (WAN)	Public	ISP (usually dynamic)	Visible to internet
Router (LAN)	Private (192.168.1.1)	Static	Default gateway
Laptop	Private (192.168.1.2)	Dynamic (DHCP)	Changes occasionally

Device	IP Type	How Assigned	Notes
Phone	Private (192.168.1.3)	Dynamic (DHCP)	Changes occasionally
Smart TV	Private (192.168.1.4)	Dynamic (DHCP)	Connected for streaming

How It Works:

1. Router gets **public IP** from ISP
2. Router uses **NAT** to share this IP with all devices
3. Each device has a **private IP** on the internal network
4. **DHCP** automatically assigns private IPs
5. Devices can communicate internally using private IPs
6. Internet requests go through NAT (private → public IP)

Scenario: Web Server

Device	IP Type	How Assigned	Notes
Web Server	Public	Static	Always reachable at the same IP
Email Server	Public	Static	Reliable for email delivery

8. Lesson Sequence

Lesson Plan

Phase	Time	Activity
Introduction	5 min	"Why do we need IP addresses?"; state learning objectives
IPv4 vs. IPv6	15 min	Explain differences, address format, address space; limitations of IPv4
Public vs. Private; Static vs. Dynamic	15 min	Explain distinctions; DHCP process; real-world examples
NAT	15 min	Role of NAT; diagram; security benefits
Activities/Worksheet	10 min	Hands-on IP discovery; quizlet; Q&A

Activity: Find Your IP Address

Activity	Description
Find Public IP	Students visit a website (e.g., whatismyip.com) to find their public IP
Find Private IP	Students find their private IP on their device (varies by OS)
Static vs. Dynamic	Students investigate whether their IP is static or dynamic
Quizlet	Students use Quizlet to reinforce vocabulary

9. Assessment Activities

Assessment Type	Description
Class Participation	Observe engagement and contributions
Worksheet Questions	Compare IPv4 and IPv6; explain public/private; describe NAT
Quiz	Multiple-choice quiz on eBook page A2.3.1
Exit Ticket	Gauge understanding and identify areas needing clarification

Sample Worksheet Questions:

1. Compare and contrast IPv4 and IPv6 in a table.
 2. Explain the difference between a public and a private IP address, and give an example of each.
 3. Explain the difference between a static and a dynamic IP address, and give an example of each.
 4. Describe how NAT works and explain its benefits.
 5. Why is the internet moving to IPv6?
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10. Differentiation

Level	Strategy
Support	Provide visual aids, concise definitions, and real-world examples

Level	Strategy
Challenge	Encourage research on IPv6 adoption and security benefits; explore advanced NAT configurations

11. Resources

- Presentation (Slide Deck)
- eBook (A2.3.1 with quiz)
- Quizlet (A2.3.1 vocabulary flashcards)
- Online Tools: WhatIsMyIP, DHCP configuration guides

12. Summary: Key Takeaways

Concept	Key Point
IPv4	32-bit; ~4.3 billion addresses; running out
IPv6	128-bit; virtually unlimited addresses; future-proof
Public IP	Globally unique; accessible over internet; assigned by ISP
Private IP	Used within private networks; not accessible over internet; assigned by router/DHCP
Static IP	Manually assigned; does not change; reliable
Dynamic IP	Automatically assigned (DHCP); may change; flexible
NAT	Translates private IPs to public IPs; conserves addresses; enhances security

"IP addresses are the foundation of internet communication—they identify devices, enable routing, and ensure data reaches the right destination."